# Week 8 — Forecasting

## Classical to Modern Forecasters

*Resource Guide & Learning Toolkit*

Statistical Machine Learning · Fusemachines AI Fellowship
Facilitator: Susan Ghimire | Dataset: Monthly S&P 500 Index 1990–2024 (yfinance)

**How to use this guide:** Organised in five sections mirroring the module phases. **Step 1** — Watch the priority videos (all under 30 min). **Step 2** — Read the corresponding FPP3 chapter — free, online, interactive. **Step 3** — Work through the Kaggle tutorials for hands-on practice. **Step 4** — Use the AI prompts: paste each into Claude or ChatGPT, then apply the critique guidance to verify the answer. **Step 5** — Bring the discussion questions to your mid-week sync. All links are clickable. Resources marked ★ **Priority** are the minimum required.

## LEARNING PHASES

| Phase | Topics |
|---|---|
| **0**<br>***Setup & Working with Financial Time-Series Data*** | • What is a time series? Trend, seasonality, cyclic, noise components<br>• Stationary vs non-stationary; EMH and random walks for financial data<br>• yfinance download; DatetimeIndex with freq='MS'; resample(), asfreq()<br>• Handling missing data: ffill, bfill, interpolate(method='time')<br>• Log prices vs log returns; ADF + KPSS stationarity tests<br>• ACF / PACF correlograms; STL decomposition |
| **1**<br>***Classical Models → SARIMA*** | • Naive (random walk) and Seasonal Naive: MASE = 1.0 baselines<br>• Holt-Winters Triple ES: $\alpha$, $\beta$, $\gamma$ (level + trend + seasonality)<br>• Box-Jenkins workflow: identify → estimate → diagnose → forecast<br>• SARIMA(p,d,q)(P,D,Q,12): reading ACF/PACF to choose orders<br>• Ljung-Box residual test; QQ-plot; np.exp() inverse-transform<br>• AutoARIMA (pmdarima) for order validation |
| **2**<br>***Automated Models + Feature Engineering*** | • Prophet: piecewise linear trend, Fourier seasonality, changepoints<br>• changepoint_prior_scale=0.5 for volatile financial data<br>• Prophet cross_validation() + performance_metrics() for TS CV<br>• Connection to W6 GP: composite kernel (trend + periodic + noise)<br>• make_features(): lag_1/2/3/6/12, roll_12_mean/std, month, year, trend (10 total)<br>• Walk-forward CV: TimeSeriesSplit (expanding window) — never standard k-fold |
| **3**<br>***Machine Learning + Deep Learning*** | • XGBoost: gradient boosting on tabular lag features; walk-forward CV<br>• LightGBM: recursive multi-step forecast with history buffer append<br>• LightGBM quantile regression: prediction intervals ($\alpha$=0.025, 0.975)<br>• MLP: sklearn MLPRegressor on same tabular features (StandardScaler required)<br>• LSTM: input shape (samples, LOOKBACK=12, 1); MinMaxScaler; recursive forecast<br>• Transformer reference: pytorch-forecasting (TFT), darts (N-BEATS, PatchTST) |
| **4**<br>***Evaluation, Error Analysis & Business Output*** | • Metrics: MAE, RMSE, MAPE, sMAPE, WMAE, WMAPE, MASE (MASE < 1.0 = beats naive)<br>• Error by calendar month; by forecast horizon h=1..H; by value bins<br>• Simple ensemble: average of 4 best models (benefit ∝ error de-correlation)<br>• Diebold-Mariano test: H■ = equal predictive accuracy; $p < 0.05$ = significant<br>• Hierarchical TS overview: Bottom-Up, Top-Down, MinT reconciliation<br>• Investment recommendation: EMH context, structural break risk, non-accuracy criteria |

# RESOURCES

## ★ Priority Videos — Watch Before the Session

### ➤ Time Series Decomposition — ritvikmath

https://youtu.be/v5ijNXvlC5A

★ Priority · Phase 0–1. Builds the core mental model of trend, seasonality, and residuals with visual examples. Covers additive vs multiplicative decomposition. **Watch this first.**

### ➤ ACF and PACF Explained — ritvikmath

https://youtu.be/DeORzP0go5I

★ Priority · Phase 1. The clearest explanation of how to read ACF and PACF plots to identify AR and MA orders. Essential before the SARIMA model identification step.

### ➤ Coding the SARIMA Model in Python — ritvikmath

https://youtu.be/Al8m6K_stfA

★ Priority · Phase 1. Full Box-Jenkins pipeline in Python: identification, estimation, diagnostics, forecast. Uses statsmodels SARIMAX. Covers seasonal differencing and SARIMA order selection.

### ➤ Holt-Winters Exponential Smoothing — Meerkat Statistics

https://youtu.be/vQCcD0j-vHQ

★ Priority · Phase 1. Explains α, β, γ smoothing parameters visually with trend and seasonality examples. Understand this before fitting ExponentialSmoothing in statsmodels.

### ➤ Time Series Forecasting with Prophet — Greg Hogg

https://youtu.be/KvLG1uTC-KU

Phase 2. Hands-on Prophet walkthrough using real stock price data (~30 min). Covers data formatting (ds/y), changepoints, uncertainty intervals, and component plots. Directly relevant to the S&P 500 assignment context.

### ➤ Feature Engineering for Time Series — Kishan Manani (PyData London 2022)

https://youtu.be/9QtL7m3YS9I

★ Priority · Phase 2–3. How to convert a time series into a lag-feature matrix for gradient boosting. Covers lag features, rolling statistics, and walk-forward validation.

### ➤ MASE vs RMSE: Forecast Evaluation — ritvikmath

https://youtu.be/V0XVqMiCGfE

Phase 7. Explains why MASE is the preferred scale-independent metric and how to compute it relative to the naïve seasonal benchmark.

## Online Courses

### ■ Udacity: Time Series Forecasting (optional)

https://www.udacity.com/course/time-series-forecasting--ud980

Optional · All phases. Covers classical models (ARIMA, ES) and ML approaches with Python. Good structured alternative to videos if you prefer a course format. Estimated 4 weeks at 5–10 hrs/week.

### ■ Coursera: TensorFlow — Sequences, Time Series & Prediction

https://www.coursera.org/learn/tensorflow-sequences-time-series-and-prediction

Phase 6. DeepLearning.AI course (Andrew Ng). Covers RNN, LSTM, CNN, and Transformer approaches to time series in TensorFlow. Prerequisite: basic TF/Keras. Take this if you want to go deeper on the deep learning phase.

### ■ Prophet Quickstart — Official Meta Docs (Python)

https://facebook.github.io/prophet/docs/quick_start.html

Phase 2. Official Prophet quickstart with worked Python examples. Covers the ds/y DataFrame format, fitting, make_future_dataframe, forecasting, and component plots. Read this before Q11–Q12. No prerequisites beyond Python.

## Textbooks & Long-Form Reading

### ● Forecasting: Principles and Practice (FPP3) — Hyndman & Athanasopoulos

https://otexts.com/fpp3/

★ Priority · All phases. Free online, interactive. The definitive open-access forecasting textbook. Key chapters: 3 (Decomposition), 5 (Exponential Smoothing), 9 (ARIMA), 11 (Hierarchical TS), 13 (Practical Issues). Read Ch. 3 before the session and Ch. 9 before SARIMA.

### ● Forecasting at Scale (Prophet paper) — Taylor & Letham 2018

https://peerj.com/preprints/3190.pdf

Phase 4. Original Prophet paper — explains piecewise linear trend, Fourier seasonality, and Bayesian changepoint prior. 15 pages, highly readable.

● **Introduction to Time Series Forecasting with Python — Jason Brownlee**

*https://machinelearningmastery.com/introduction-to-time-series-forecasting-python/*

Phases 0–5. Practical recipes for loading, visualising, transforming, and fitting classical models (ARIMA, ES) in Python. Best used as a hands-on reference alongside FPP3. Code-first approach.

● **Practical Time Series Analysis — Aileen Nielsen**

*https://www.oreilly.com/library/view/practical-time-series/9781492041641/*

All phases. Comprehensive coverage of statistics and ML approaches. Strong on evaluation and error analysis (Section 4). O'Reilly — check library access.

● **Deep Learning for Time Series Forecasting — Jason Brownlee**

*https://machinelearningmastery.com/deep-learning-for-time-series-forecasting/*

Phase 6. Practical coverage of MLP, CNN, and LSTM models in Keras/Python. Each chapter includes working code. Use for Phase 6 deep dive.

## Kaggle Tutorials — Hands-On Practice

◆ **Kaggle Learn: Time Series**

*https://www.kaggle.com/learn/time-series*

★ Priority · All phases. Kaggle's official time series course — 4 lessons covering trend/seasonality, lag features, walk-forward CV, and hybrid models. Includes executable notebooks. **Complete Lessons 1–3 before the session.**

◆ **Topic 9: Time Series Analysis in Python — Kashnitsky (mlcourse.ai)**

*https://www.kaggle.com/kashnitsky/topic-9-part-1-time-series-analysis-in-python*

Phases 1–3. Excellent notebook covering stationarity, ACF/PACF, ARIMA, and feature engineering for ML. Dense and technical — skip to the section you need.

◆ **Time Series Analysis: A Complete Guide — Andreshg**

*https://www.kaggle.com/andreshg/timeseries-analysis-a-complete-guide*

All phases. Wide-coverage notebook from EDA through ARIMA, Prophet, and LSTM. Good for seeing all techniques on one dataset before implementing your own.

◆ **Everything You Can Do with a Time Series — TheBrownViking20**

*https://www.kaggle.com/thebrownviking20/everything-you-can-do-with-a-time-series*

Phases 0–4. Broad survey notebook: decomposition, stationarity, classical models, visualisation. Good companion for Phase 0–1 exploration.

## Kaggle Competition Datasets — For Extension Projects

**Extension dataset guidance:** The assignment uses yfinance to download S&P 500 monthly data (internet required; a synthetic fallback is built in). These Kaggle datasets are for independent projects and further practice. Each tests a different forecasting challenge (hierarchical, multiple seasonality, intermittent demand). A Kaggle account is required.

■ **Recruit Restaurant Visitor Forecasting**

*https://www.kaggle.com/c/recruit-restaurant-visitor-forecasting*

Extension. Hierarchical daily restaurant visitor counts across Japan. Tests: hierarchical TS, multiple seasonalities (weekly + annual), cold-start problem (new restaurants). Excellent for Phase 7 hierarchical extension.

■ **Web Traffic Time Series Forecasting**

*https://www.kaggle.com/c/web-traffic-time-series-forecasting*

Extension. 145,000 Wikipedia page daily views — multivariate, high-dimensional. Tests: scaling forecasting to many series, missing data, LSTM approaches. Good for Phase 6 (deep learning) extension.

### ■ Rossmann Store Sales

https://www.kaggle.com/c/rossmann-store-sales

Extension. 1,115 German drug stores, daily sales with exogenous variables (promotions, holidays, school holidays). Tests: exogenous feature engineering (Phase 5), hierarchical aggregation.

### ■ Property Sales

https://www.kaggle.com/htagholdings/property-sales

Extension. Monthly property sales in Australia by region. Tests: spatial hierarchy, trend and seasonal decomposition, STL.

## Official Documentation

### § statsmodels: SARIMAX State Space Models

https://www.statsmodels.org/stable/statespace.html

Phase 3 — Essential. API reference for SARIMAX, model summary, get_forecast(), conf_int(), and residual diagnostics including acorr_ljungbox().

### § statsmodels: Exponential Smoothing (Holt-Winters)

https://www.statsmodels.org/stable/examples/notebooks/generated/exponential_smoothing.html

Phase 2. Tutorial notebook for SimpleExpSmoothing, Holt, and ExponentialSmoothing. Includes the airline passengers example — directly applicable.

### § Prophet Documentation — Meta

https://facebook.github.io/prophet/

Phase 4. Official Prophet docs with Python examples. Read 'Quick Start' and 'Cross Validation' before Q15–Q16.

### § pmdarima: auto_arima Example

https://alkaline-ml.com/pmdarima/auto_examples/arima/example_auto_arima.html

Phase 3. AutoARIMA API reference — automated order selection. Useful for Q14 (optional) to validate your manual ACF/PACF identification.

### § LightGBM Documentation

https://lightgbm.readthedocs.io/en/stable/

Phase 5. API reference. Key: quantile regression objective (objective='quantile', alpha=0.025/0.975) for prediction intervals.

### § pytorch-forecasting: TFT, N-BEATS

https://pytorch-forecasting.readthedocs.io/en/stable/

Phase 6 (reference). Implements Temporal Fusion Transformer and N-BEATS in PyTorch. For when you want to go beyond LSTM on larger datasets.

### § darts: Time Series Library

https://unit8co.github.io/darts/

All phases (reference). Unified interface for classical, ML, and DL forecasting models. Useful for comparing many models quickly; supports backtesting and ensemble.

# AI PROMPTS & CRITIQUE GUIDE

**How to use these prompts:** Paste each prompt into Claude or ChatGPT. Apply the **Critique / Verify** guidance. If the model makes an error, force it to fix it. Write a short note in your notebook about what each model got right and wrong — this is itself a valuable learning exercise.

## Phase 0 — Stationarity: Prices vs Returns

*Ask: "Why are stock prices non-stationary but log-returns approximately stationary? Show me ADF and KPSS tests in Python on monthly S&P; 500 data."*

**Critique / Verify:** Must explain that prices have a **stochastic trend (unit root)** — each value depends on the previous. Must note log-returns approximate stationarity (constant mean/variance). Must run both ADF (H■: unit root) and KPSS (H■: stationary) and explain what conflicting results mean. If it only runs ADF, push back: ask why running only one test can be misleading.

## Phase 1 — SARIMA residual diagnostics

*Ask: "Write Python code to fit a SARIMA model on monthly S&P; 500 log-prices, run residual diagnostics, and forecast 12 steps ahead."*

**Critique / Verify:** Must use **SARIMAX from statsmodels**. Must include Ljung-Box test and QQ-plot. Must apply **np.exp()** to inverse-transform back to prices. Ask: 'What happens if Ljung-Box $p < 0.05$?' — if it says 'ignore it', push back. Ask: 'For near-white-noise returns, what ARIMA order would be equivalent to a random walk?' — answer should be ARIMA(0,1,0).

## Phase 2 — Prophet changepoints for financial data

*Ask: "How does changepoint_prior_scale affect Prophet forecasts on volatile financial data? Why use 0.5 instead of the default 0.05?"*

**Critique / Verify:** Must explain changepoints as locations where **growth rate changes** — not trend level jumps. Must correctly state: higher scale = more flexible trend = more changepoints detected. Must connect to the sparse Laplace prior. For financial data, must explain why the default 0.05 produces an overly rigid trend. If it just says 'tune this hyperparameter', push back — ask for the prior.

## Phase 2–3 — Walk-forward CV for financial forecasting

*Ask: "Why is standard k-fold cross-validation wrong for time series? Show the correct approach with TimeSeriesSplit on stock price data."*

**Critique / Verify:** Must state that k-fold **shuffles the index**, leaking future observations into training. Must name the fix: **TimeSeriesSplit (expanding window)**. Must explain that for financial data, leakage is especially problematic — training on 2020 data to predict 2019 is economically nonsensical. If it only says 'use time-based split' without the leakage mechanism, push back.

## Phase 3 — LSTM input shape and recursive forecast

*Ask: "Show me how to build a Keras LSTM for 48-step-ahead monthly financial forecasting. What is the correct input shape and how do I handle MinMaxScaler?"*

**Critique / Verify:** Must state input shape: **(samples, LOOKBACK, 1)** — a 3D tensor. Must apply MinMaxScaler to log-prices before training. Must implement a **recursive forecast loop**: append each prediction to the window before predicting the next step. Must inverse_transform at the end. If it only trains one-step-ahead, ask how to extend to 48 steps.

### Phase 4 — WMAPE vs MAPE for investment forecasting

*Ask: "What is the difference between MAPE and WMAPE? When does WMAPE give a more useful result for investment portfolio forecasting?"*

**Critique / Verify:** Must explain that WMAPE weights each error by **true value** — errors at higher index levels (e.g., S&P 500 at 4500) count more than errors at lower levels (e.g., 1500). Must give the formula: $\Sigma(|e/y| \times y) / \Sigma(y) \times 100$. For finance: a 10% miss on a $10M position is more costly than a 10% miss on a $1M position — WMAPE captures this. If it conflates WMAPE with sMAPE, correct it.

## MID-WEEK DISCUSSION QUESTIONS

**How to use:** Write a 2–3 sentence answer in your notebook before the session. During discussion, compare reasoning with peers. The quality of your justification matters more than the conclusion.

### Q1. Random Walk vs Forecastable

Your ACF/PACF of S&P 500 log-returns shows no statistically significant lags. Does this mean all forecasting models will produce MASE ≈ 1.0? If so, is building a forecasting pipeline worthless? What value does a well-calibrated uncertainty interval (prediction band) provide even when point forecasts cannot beat a random walk?

### Q2. COVID Structural Break

Your SARIMA forecast from December 2019 predicts a steady upward trend, but the March 2020 actual value is –34% vs the forecast. Is this a model failure or working as expected? How should a quant analyst communicate forecasting limitations around tail risks to an investment committee that demands 12-month point forecasts?

### Q3. Ljung-Box Failure at Lag 12

Your SARIMA(1,1,1)(0,1,1,12) passes Ljung-Box at lags 6 and 24 but fails at lag 12 (p=0.03). What does this indicate? Which specific parameter(s) would you adjust, and how would you verify the fix?

### Q4. LightGBM vs SARIMA: Statistical Significance

Your LightGBM achieves MASE=0.87 vs SARIMA MASE=0.91, but the Diebold-Mariano p-value is 0.18. Does the better MASE 'count'? What investment decision risk comes from acting on a performance gap that is not statistically significant?

### Q5. Recursive vs Direct Multi-step

Recursive forecasting feeds each prediction back as a lag feature for the next step. Direct forecasting trains one model per horizon h=1,2,...,48. At what horizon does error compounding make recursive forecasting unreliable? When does direct forecasting degrade more slowly?

### Q6. Hierarchical Portfolio Forecasting

Your firm manages a portfolio of 50 S&P 500 stocks. Option A: forecast each stock, then sum. Option B: forecast the index, then disaggregate. Which gives better portfolio-level accuracy? Which gives better stock-level accuracy? What does Optimal Reconciliation (MinT) add that neither method can achieve alone?

### Q7. Quantile Forecasts: 74% Coverage

Your LightGBM 95% prediction interval achieves only 74% empirical coverage on the 2020–2024 test set. Name two possible causes. How would you diagnose whether the shortfall is due to model miscalibration (training issue) vs distributional shift (COVID changed the data-generating process)?

### Q8. Deployment Decision: Non-Accuracy Criteria

Three models have nearly identical MASE (0.88, 0.89, 0.90) on the S&P 500 test. List 3 non-accuracy criteria for choosing between them and rank them for this specific use case: monthly equity allocation, 12-month horizon, CIO reports to investment regulators who require model explainability.

## PHASE → RESOURCE MAP

| Phase | Topic | ★ Watch / Practice | ★ Read |
|---|---|---|---|
| 0 | Data Prep<br>Stationarity<br>ACF/PACF | Decomp — ritvikmath<br>ACF/PACF — ritvikmath<br>Kaggle Learn: TS | FPP3 Ch. 1–2<br>FPP3 Ch. 3 |
| 1 | Classical<br>SARIMA | Holt-Winters — Meerkat Stats<br>SARIMA Coding — ritvikmath | FPP3 Ch. 5<br>FPP3 Ch. 9<br>statsmodels SARIMAX |
| 2 | Prophet<br>Feature Eng. | Prophet — Greg Hogg<br>TS Features — Manani | Prophet Docs<br>Prophet Quickstart |

| Phase | Topic | ★ Watch / Practice | ★ Read |
|---|---|---|---|
| 3 | XGBoost<br>LightGBM<br>LSTM | Coursera TF Sequences<br>Kaggle Topic 9 | LightGBM Docs<br>Brownlee: DL for TS |
| 4 | Evaluation<br>Ensemble<br>Hierarchical | MASE — ritvikmath | FPP3 Ch. 5: Accuracy<br>FPP3 Ch. 11: Hierarchical |